

From Fragmented Signals to Timely Student Support

- A governed student-retention workflow on Databricks
- Public Sector • Higher Education • Synthetic demonstration

Advisors need to know who needs attention today—and why—without compromising trust.

Buyer KPIs and synthetic demo baseline

- Designed values demonstrate the workflow—not real institutional performance.
- 88.88%
- Synthetic retention rate
- 2,664
- Initially elevated-risk students
- \$24.1M
- Estimated synthetic tuition exposure
- 20,000
- Current synthetic students

Current-state friction delays action

- The problem is operational, not simply predictive.
- Fragmented signals
- Attendance, LMS, finance, enrollment, and case data arrive on different clocks.
- Inconsistent prioritization
- Advisors reconcile data manually and lack a defensible daily queue.
- Weak feedback loop
- Outreach and outcomes are disconnected from executive coverage and analytics.

A closed-loop workflow from signal to action

- One governed path connects observation, human action, and analytical learning.
- 1. Observe — trigger one deterministic synthetic day
- 2. Govern — apply quality, lineage, grants, and row filters
- 3. Prioritize — score current snapshots with a versioned ML model
- 4. Assist — generate cited, evaluated advisor briefings
- 5. Act — record interventions transactionally in Lakebase
- 6. Learn — return CDC to governed Gold
- 7. Understand — ask embedded Genie and inspect its SQL

Business value for two accountable owners

- A single experience serves different decisions without collapsing access boundaries.
- Executive sponsor
- See aggregate risk concentration, intervention coverage, retention, and estimated tuition exposure—without student detail.
- Student-success owner
- Reduce signal-to-outreach time with a prioritized caseload, grounded context, and closed-loop follow-up history.

Governed Databricks architecture

- Serving refreshes cannot overwrite advisor transactions.
- Synthetic signals → Lakeflow Bronze / Silver → MLflow scoring → Unity Catalog Gold → read-only Lakebase serving → Databricks App + embedded Genie → Lakebase-owned interventions → CDC → governed intervention Gold

Hybrid intelligence keeps each decision explainable

- Use each capability for what it does best.
- ML classifier
- Repeatability, prioritization, calibrated scores, stable tiers, version history, and baseline comparison.
- Grounded GenAI
- Allowlisted facts, citations, bounded discussion prompts, and fixed-set evaluation.
- Human advisor
- Judgment, outreach, reconciliation, and accountable action remain with people.

Live demo: one coherent student-success story

- Expected duration: 12–15 minutes; no manual data repair.
- Replay the September 23 synthetic incident day
- Inspect quality and feature freshness
- Show a medium-to-high risk transition
- Review the governed caseload and cited briefing
- Record outreach and a follow-up
- Verify Lakebase CDC and intervention Gold
- View aggregate executive coverage
- Ask embedded Genie and inspect generated SQL

Execution evidence—not screenshots

- Every build domain has committed, text-readable proof.
- 60,000
- Immutable daily risk-history rows
- 0.9669
- Synthetic validation PR-AUC
- SUCCESS
- Workflow and idempotent replay
- ONLINE
- Both serving sync tables

Trust is designed in—and limitations stay visible

- Responsible use means exposing both controls and boundaries.
- Trust controls
 - Row-filtered advisor access
 - Aggregate-only executive products
 - Read-only serving tables
 - Cited AI and inspectable SQL
 - Visible stale, partial, and error states
- Honest boundaries
 - Synthetic data and designed signals
 - No causal impact claim
 - No automated adverse action
 - No institutional validation yet
 - Triggered daily—not real time

Pilot the workflow—not just the model

- Align owners • validate institutional data and thresholds • measure signal-to-outreach time and intervention coverage
- Governed decisions. Accountable action. Measurable learning.